

Peroxisome Biogenesis Disorder 6A (Zellweger) — Comprehensive Disease Characteristics Report

Disease: Peroxisome Biogenesis Disorder 6A (Zellweger) — PBD6A **MONDO ID:** MONDO:0013936 · **OMIM:** #614870 (PBD6A) / #614871 (PBD6B, milder allelic form) · **Gene:** PEX10 (OMIM 602859, 1p36.32) **Category:** Mendelian, autosomal recessive

Summary

Peroxisome Biogenesis Disorder 6A (Zellweger) is the **severe, PEX10-caused end of the Zellweger spectrum disorder (ZSD) continuum** — a fatal, autosomal-recessive inborn error of organelle biogenesis. Biallelic loss-of-function mutations in **PEX10**, one of ~13–14 *PEX* genes required for peroxisome assembly, disable the peroxisomal matrix-protein import machinery. PEX10 is a **C3HC4 zinc-RING-finger peroxin** that, together with PEX2 and PEX12, forms the E3-ubiquitin-ligase core of the peroxisomal "importomer." This ligase mono- and poly-ubiquitinates the cycling receptor PEX5 to drive its recycling; when it fails, PEX5-mediated delivery of matrix enzymes collapses and peroxisomes become **functionally empty ghosts**. RING-finger missense substitutions (e.g., p.Cys307Tyr, p.Cys276Phe, p.Arg311Gln) map to this critical domain, and the most severe (Zellweger) phenotypes arise when residual function is minimal ([PMID: 32069232](#), [PMID: 28320181](#), [PMID: 20679226](#)).

The downstream biochemistry is a **multi-pathway metabolic failure**: impaired α - and β -oxidation of very-long-chain fatty acids (VLCFA), defective bile-acid synthesis with accumulation of toxic C27 intermediates (DHCA, THCA), and deficient ether-phospholipid (plasmalogen) and DHA synthesis ([PMID: 34625341](#)). These lesions converge on **mitochondria-mediated oxidative stress and neuronal death**, producing the hallmark neuronal-migration defects (polymicrogyria, pachygyria, germinolytic cysts) alongside hepatic, renal, retinal, auditory, adrenal and skeletal disease ([PMID: 20959636](#)). Classic Zellweger neonates present at birth with profound hypotonia, seizures, craniofacial dysmorphism and failure to thrive, and **most die within the first year** ([PMID: 28784167](#), [PMID: 12069541](#)).

ZSD is rare (~1 in 50,000–90,000 births, with regional founder effects), diagnosed by **elevated plasma VLCFA (C26:0, C26:0/C22:0) and the dried-blood-spot marker C26:0-lysophosphatidylcholine** confirmed by *PEX*-gene sequencing (PMID: 28677031, PMID: 30846882). Management is **supportive**; oral **cholic acid (Cholbam®)** is the sole FDA-approved therapy, and gene/base-editing approaches show preclinical rescue in mouse models (PMID: 34521419, PMID: 41981313). Prevention is limited to genetic counseling, carrier testing and prenatal diagnosis (PMID: 23327810).

1. Disease Information

Overview. Zellweger syndrome (ZS), historically the "**cerebro-hepato-renal syndrome**," was first described in 1964 as a familial syndrome of multiple congenital defects, and is the **severest** of the peroxisome biogenesis disorders (PBDs) (PMID: 12069541, PMID: 23327810). PBD6A is the specific designation for the **PEX10-caused severe (Zellweger) form**; PBD6B is the milder PEX10 form (neonatal adrenoleukodystrophy / infantile-Refsum-like). The three classic clinical labels — Zellweger syndrome (severe), neonatal adrenoleukodystrophy (moderate), and infantile Refsum disease / ataxic form (mild) — are now recognized as a **continuum, the Zellweger spectrum disorder (ZSD)** (PMID: 28320181).

Key identifiers.

Resource	Identifier
OMIM	#614870 (PBD6A) · #614871 (PBD6B) · 602859 (PEX10 gene)
MONDO	MONDO:0013936 (PBD6A)
Orphanet	ORPHA:912 (Zellweger syndrome)
MeSH	D015211 (Zellweger Syndrome)
ICD-10	Q87.8 / E71.5x (peroxisomal disorders)
ICD-11	5C57.0

Synonyms / alternative names. Zellweger syndrome; cerebro-hepato-renal syndrome (CHRS); PBD, Zellweger type; peroxisome biogenesis disorder 6A; PEX10-related ZSD (PMID: 23327810, PMID: 12069541).

Information source. Content is drawn from **aggregated disease-level resources** (OMIM, Orphanet, natural-history cohorts, scoping reviews) supplemented by **individual patient case reports** and small clinical cohorts; there is no large EHR-derived dataset given rarity.

2. Etiology

Primary cause — genetic. PBD6A is caused by **biallelic (homozygous or compound-heterozygous) pathogenic variants in PEX10** (PMID: 32069232). "Mutations of 13 different PEX genes lead to PBDs including Zellweger syndrome (ZS)" and "different types of mutations of PEX1 and PEX10 genes are correlated with broad-range phenotypes of PBDs" (PMID: 32069232). There is **no environmental or infectious cause**; the disease is fully genetically determined.

Genetic risk factors. The causal factor is the loss-of-function *PEX10* genotype itself. **Genotype–phenotype correlation** is central: RING-domain missense changes and severe truncating alleles yield Zellweger; hypomorphic/milder alleles yield NALD or the ataxic form (PMID: 28320181, PMID: 27230853). **Consanguinity** and membership in **founder populations** (see §9) increase risk.

Environmental / lifestyle / infectious factors. None causative. This is not a multifactorial or exposure-driven disease. Infections (e.g., neonatal sepsis) are **complications**, not causes (PMID: 33213396).

Protective factors. No established genetic or environmental protective factors. Within the spectrum, **residual PEX10 function (hypomorphic alleles)** is the principal modifier that "protects" against the severe Zellweger phenotype, shifting patients toward milder, longer-surviving disease (PMID: 27230853).

Gene–environment interactions. Not applicable in a conventional sense; phenotype is modulated chiefly by allele severity and, in prolonged-survival cases, likely by unknown modifier factors (PMID: 15098231).

3. Phenotypes

Classic (severe) Zellweger presents in the **neonatal period** with a stereotyped, multisystem, progressive phenotype. A scoping review/meta-analysis (107 studies, 307 patients) and a 136-patient natural-history chart review found clinical findings differing significantly across severity categories ([PMID: 35741019](#)).

Phenotype	Type	HPO term	Onset	Frequency / severity
Severe hypotonia	Clinical sign	HP:0001319 (neonatal hypotonia)	Neonatal	Near-universal, severe
Seizures	Clinical sign	HP:0001250	Neonatal	Very common
Craniofacial dysmorphism (high forehead, large fontanelle, flat face, epicanthus, broad nasal bridge, micrognathia)	Physical	HP:0000280 / HP:0000239	Congenital	Characteristic
Retinal degeneration → blindness	Lab/clinical	HP:0000546	Infancy	"Almost all" (PMID: 37541626); median VA ~0.93 logMAR (~20/320)
Sensorineural hearing loss	Clinical	HP:0000407	Infancy	Moderately-severe to severe, slowly progressive (PMID: 34534157)
Hepatic dysfunction (hepatomegaly, cholestasis, fibrosis, coagulopathy)	Lab/clinical	HP:0002240 / HP:0001394	Neonatal	Common
Renal cysts, hyperoxaluria/stones	Imaging/lab	HP:0000107	Congenital	Common
Adrenal insufficiency	Lab	HP:0000846	Variable	Reported
Global developmental delay / no milestones	Behavioral/developmental	HP:0001263	Infancy	Severe form reaches no milestones (PMID: 28784167)
	Clinical		Neonatal	Common

Phenotype	Type	HPO term	Onset	Frequency / severity
Feeding difficulties / failure to thrive / GERD		HP:0011968 / HP:0001508		
Chondrodysplasia punctata / epiphyseal stippling, fractures	Imaging	HP:0000943	Congenital	Reported

"Common clinical findings that were significantly different across severity categories included seizures, hypotonia, reduced mobility, feeding difficulties, renal cysts, adrenal insufficiency, hearing and vision loss, and a shortened lifespan" ([PMID: 35741019](#)).

Quality-of-life impact. In severe Zellweger, QoL is profoundly limited: affected infants reach no developmental milestones, are cortically blind and deaf, feed poorly and rarely survive infancy ([PMID: 28784167](#)). Milder spectrum survivors may achieve supported employment and partly independent living, but face progressive gait disorders and sensory loss ([PMID: 26287655](#)).

4. Genetic / Molecular Information

Causal gene. **PEX10** (HGNC gene, OMIM 602859), chromosome **1p36.32**, encoding an integral peroxisomal-membrane peroxin. PEX10 is "involved in the import of peroxisomal matrix proteins, and the mutation of this gene causes 3 subtypes of peroxisome biogenesis disorders, namely Zellweger syndrome (severe), neonatal adrenoleukodystrophy (moderate) and an ataxic form (mild)" ([PMID: 28320181](#)).

Pathogenic variants. - **Variant types:** missense (esp. RING-domain), frameshift, nonsense/truncating, and small deletions. Representative pathogenic changes: **p.Cys307Tyr (p.C307Y)** in the RING finger ([PMID: 28320181](#)); **p.Cys276Phe** and **p.Arg311Gln** in the ataxic form ([PMID: 27230853](#)); a **homozygous 2-bp deletion** as a Japanese founder allele ([PMID: 12794690](#)). - **Classification (ACMG/AMP):** null/frameshift/nonsense alleles are pathogenic (PVS1); RING-domain missense in a well-established functional domain are typically pathogenic/likely pathogenic. Milder cases may carry one hypomorphic allele. - **Allele frequency:** individually very rare in gnomAD; carrier frequencies elevated in founder populations (see §9). - **Origin:**

germline; no somatic relevance. - **Functional consequence: loss of function** (impaired E3-ligase/importomer activity). No gain-of-function or dominant-negative mechanism is described ([PMID: 20679226](#)).

Modifier genes. Allele severity is the dominant modifier; unidentified factors influence phenotype even for identical genotypes ("next to the PEX1 genotype other yet unknown factors determine the ultimate phenotype," [PMID: 15098231](#) — an observation that generalizes across ZSD).

Epigenetic information. No disease-specific DNA-methylation or histone-modification signatures have been established for PBD6A.

Chromosomal abnormalities. None characteristic; PBD6A is a single-gene disorder, not a copy-number/structural syndrome.

5. Environmental Information

Environmental factors: none causative — PBD6A is a purely genetic disorder. **Lifestyle factors:** not applicable. **Infectious agents:** not causative; however, neonates are vulnerable to **overwhelming Gram-negative sepsis** as a complication, consistent with an emerging role of peroxisomes in immune modulation ([PMID: 33213396](#)). Dietary considerations (VLCFA restriction, DHA and fat-soluble-vitamin supplementation) are therapeutic/supportive rather than etiologic.

6. Mechanism / Pathophysiology

Ordered causal chain

1. **Biallelic PEX10 loss-of-function mutation** (esp. C3HC4 RING-finger substitution) **leads to** loss of PEX10's E3-ubiquitin-ligase activity within the peroxisomal importomer ([PMID: 20679226](#)).
2. This **results in** failure to mono-/poly-ubiquitinate the cycling PTS1 receptor **PEX5** (and PTS2 co-receptor PEX7/PEX20), blocking receptor recycling ([PMID: 23344950](#)).

3. Blocked recycling **leads to** collapse of **peroxisomal matrix-protein import** — peroxisomes form membrane "ghosts" lacking matrix enzymes ([PMID: 28320181](#)).
4. Absent matrix enzymes **result in** multi-pathway metabolic failure: (a) impaired α/β -oxidation → **VLCFA accumulation (C26:0)**; (b) defective bile-acid synthesis → **toxic C27 intermediates (DHCA, THCA)**; (c) deficient ether-lipid synthesis → **plasmalogen deficiency**; (d) reduced DHA; branched-chain (phytanic/pristanic) and pipecolic-acid accumulation ([PMID: 34625341](#)).
5. These lesions **branch** into tissue injury:
6. **Brain branch:** plasmalogen deficiency + lipotoxicity **leads to mitochondria-mediated oxidative stress** ($\uparrow$ ROS, $\uparrow$ MnSOD/SOD2), neuronal apoptosis, and impaired neuronal migration → polymicrogyria/pachygyria, germinolytic cysts, gliosis ([PMID: 20959636](#)).
7. **Liver branch:** toxic C27 bile acids + VLCFA **result in** cholestasis, fibrosis/cirrhosis, coagulopathy.
8. **Kidney / eye / ear / adrenal / bone branches:** metabolic injury **results in** renal cysts, retinal degeneration, sensorineural hearing loss, adrenal insufficiency, chondrodysplasia punctata.
9. Combined multisystem failure **leads to** the classic Zellweger clinical picture and **early death, usually within the first year** ([PMID: 12069541](#)).

(Steps 1–4 are mechanistically demonstrated in cell/animal models; the oxidative-stress neurodegeneration step is directly demonstrated in a PEX13 brain model and inferred to generalize to PEX10.)

Detail by category

- **Molecular pathways:** peroxisomal matrix-protein import (importomer / receptor-recycling); ubiquitin–proteasome-linked receptor cycling. "The integral peroxisomal membrane proteins PEX10, PEX2, and PEX12 contain a zinc RING finger close to the C terminus" and act as E3 ligases for PEX5/Pex20 ubiquitination ([PMID: 20679226](#), [PMID: 23344950](#)).
- **Cellular processes:** apoptosis, oxidative-stress response, gliosis (astro-/microgliosis), defective neuronal migration; dysregulated pexophagy (PEX13/PEX5-ubiquitin axis) ([PMID: 20959636](#), [PMID: 36541703](#)).
- **Protein dysfunction:** loss of PEX10 RING-finger E3-ligase function; Zn^{2+} -coordination disruption abolishes activity (embryo-lethal in *Arabidopsis*) ([PMID: 12883010](#), [PMID: 20679226](#)).

- **Metabolic changes:** VLCFA ↑, C27 bile-acid intermediates ↑, plasmalogens ↓, DHA ↓, phytanic/pristanic/pipecolic acid ↑ (PMID: 34625341).
- **Tissue damage:** oxidative stress, mitochondrial dysfunction, apoptosis (PMID: 20959636).
- **Immune involvement:** peroxisomes modulate immune response/inflammation; deficiency associated with sepsis vulnerability (PMID: 33213396).
- **Molecular profiling (models):** Pex11α-KO mice show serum/liver/heart lipidomic, metabolomic and proteomic dysregulation (PMID: 35083512); base editing normalizes liver transcriptomes and eliminates VLCFA/BCFA/C27 bile-acid accumulation (PMID: 41981313).

Suggested ontology terms. GO:0016558 (protein import into peroxisome matrix); GO:0007031 (peroxisome organization); GO:0006635 (fatty-acid β -oxidation); GO:0008203 (cholesterol/bile-acid metabolism); GO:0006979 (response to oxidative stress); GO:0001764 (neuron migration). CL:0000540 (neuron); CL:0000121 (Purkinje cell); CL:0000573 (retinal photoreceptor); CL:0000182 (hepatocyte). CHEBI:76724 (very-long-chain fatty acid); CHEBI:36021 (plasmalogen); CHEBI:3098 (bile acid).

7. Anatomical Structures Affected

Organ / body-system level. Multisystem — the name *cerebro-hepato-renal* enumerates the three primary organs. Primary: **brain/CNS** (nervous system), **liver** (digestive/hepatobiliary), **kidney** (urinary). Secondary/associated: **eyes** (retina, lens, optic nerve), **ears** (cochlea), **adrenal glands** (endocrine), **skeleton**, **GI tract** (PMID: 37144748, PMID: 33213396).

Tissue / cell level. Neurons and neuronal-migration units (cortex, cerebellar granule and Purkinje cells), hepatocytes and biliary epithelium, renal tubular/cortical tissue, retinal photoreceptors and RPE, cochlear sensory cells. The PEX13 brain model shows "impaired cerebellar fissure/cortical layer formation, defective granule cell migration and Purkinje cell layer development" (PMID: 20959636).

Subcellular level. The **peroxisome** (GO:0005777) is absent/reduced (membrane ghosts persist); secondary **mitochondrial** dysfunction (GO:0005739) contributes to oxidative injury.

Localization / lateralization. Bilateral, symmetric multisystem involvement; brain malformations (polymicrogyria, pachygyria, germinolytic subependymal cysts) are typically bilateral. Suggested UBERON terms: UBERON:0000955 (brain), UBERON:0002037 (cerebellum), UBERON:0002107 (liver), UBERON:0002113 (kidney), UBERON:0000966 (retina), UBERON:0001846 (inner ear), UBERON:0002369 (adrenal gland).

8. Temporal Development

Onset. Congenital / neonatal. Severe Zellweger presents at or shortly after birth with hypotonia and seizures (PMID: 28784167); the pattern is chronic-progressive from birth.

Progression. In the severe form the course is **rapidly progressive**, with failure to thrive and early death, usually **before age 1 year**; patients reach no developmental milestones (PMID: 12069541). Across the broader spectrum, intermediate/mild patients survive into childhood or adulthood (cohorts to 24–35 years) with variable courses — stable, slowly declining, or with adolescent-onset progressive gait disorder/leukodystrophy (PMID: 26287655, PMID: 15098231).

Patterns. No spontaneous remission. **Critical periods:** the neonatal window is the period of greatest vulnerability and the target window for any disease-modifying intervention; leukoencephalopathy accrues with age in longer survivors (PMID: 18415699, PMID: 14872027).

9. Inheritance and Population

Epidemiology. ZSD incidence ~1 in 50,000–90,000 births. New-York newborn screening estimated ~1 in 90,000 (from 1.08 million screenings) — "Our results are close to current newborn screening estimates in New York of 1 in 90,000 births, estimated from 1.08 million screenings"; an ExAC-based bioinformatic estimate gave ~1 in 83,841 (PMID: 30846882). Japan overall ~1 in 500,000–800,000, but Okinawa 1 in 30,000 (PMID: 8914632).

Inheritance. Autosomal recessive, complete penetrance for the biochemical/genetic phenotype; **variable expressivity** governed by allele severity. No anticipation (not a repeat-expansion disorder). Germline mosaicism not a notable feature.

Founder effects / consanguinity. A **PEX6 founder mutation** in Saguenay-Lac-St-Jean, Quebec gives "Incidence of ZS was estimated to 1 in 12,191 live births, with a carrier frequency of 1 in 55" (PMID: 22894767). A **homozygous 2-bp PEX10 deletion is a founder allele** among Japanese complementation-group-B patients — "All the 11 ZS patients with group-B PBD had a common mutation, i.e., a homozygous 2-base-pair deletion in PEX10" (PMID: 12794690). Consanguinity increases risk.

Population demographics. Panethnic; regional clustering where founder alleles exist (French-Canadian Quebec; Okinawa). **Sex ratio** ~1:1 (autosomal). Age distribution skews to neonates/infants for the severe form.

10. Diagnostics

Biochemical testing (first-line). Elevated plasma VLCFA — **C26:0, C26:0/C22:0 and C24:0/C22:0 ratios**; elevated pipecolic, phytanic/pristanic acids; abnormal C27 bile-acid intermediates (DHCA/THCA); reduced erythrocyte plasmalogens ([PMID: 34625341](#)).

Dried-blood-spot marker. C26:0-lysophosphatidylcholine (C26:0-lysoPC) is sensitive: "Elevated C26:0-lysoPC levels (>72 nmol/L) were found in 86/91 ZSD DBS (n=33/37 patients) corresponding to a sensitivity of 89.2%" (median 567 nmol/L), whereas C26:0-carnitine is less sensitive (55.2%) — "C26:0-lysoPC in DBS is a sensitive and useful marker for VLCFA accumulation in patients with a ZSD" ([PMID: 28677031](#)). This is the analyte used in tandem-MS newborn screening (implemented primarily for X-ALD/ABCD1), which **incidentally detects ZSD** ([PMID: 36256460](#)).

Genetic testing (confirmatory). Sequencing the ~13 *PEX* genes — whole-exome sequencing or targeted **PBD/peroxisomal gene panels**; **single-gene PEX10 testing** where a founder allele is known. Classic functional confirmation is **complementation analysis in cultured fibroblasts**. Note that milder alleles may yield near-normal fibroblast studies, making **molecular analysis essential** at the mild end ([PMID: 19127411](#)).

Imaging. Brain MRI is highly informative: "cMRI pathology in ZSS consists of abnormal gyration pattern including polymicrogyria and pachygyria, leukencephalopathy, germinolytic cysts and heterotopias" ([PMID: 18415699](#)); polymicrogyria/pachygyria predominate in severe disease, leukoencephalopathy in longer survivors ([PMID: 14872027](#)).

Differential diagnosis. Other *PEX*-gene ZSDs (*PEX1/PEX6/PEX2/PEX12/PEX26*); **single peroxisomal enzyme defects that are "Zellweger-like"** — notably **D-bifunctional protein (HSD17B4) deficiency** ("Peroxisomal D-bifunctional protein (DBP) deficiency is an autosomal recessive disorder historically described as a Zellweger-like syndrome comprising neonatal seizures, retinopathy, hearing loss, dysmorphic features," [PMID: 32904102](#)) and acyl-CoA oxidase deficiency; rhizomelic chondrodysplasia punctata; Heimler syndrome (mild *PEX1*/

PEX6, [PMID: 26387595](#)); and non-peroxisomal causes of neonatal hypotonia/renal cysts/epiphyseal stippling such as Smith-Lemli-Opitz and warfarin embryopathy ([PMID: 40995270](#), [PMID: 39359950](#)).

Screening. Cascade carrier testing in families; prenatal diagnosis on CVS/amniocytes (VLCFA/DHAPAT enzyme assay or molecular testing); newborn C26:0-LPC screening detects ZSD as a secondary finding.

11. Outcome / Prognosis

Survival. Severity-dependent. Classic (severe) Zellweger — the cerebro-hepato-renal syndrome — "is characterized by the presence of dysmorphias and polymalformative syndrome, severe neurologic abnormalities including neurosensory defects and hepato-intestinal dysfunction with failure to thrive and usually early death," typically **before 1 year** ([PMID: 12069541](#)). Intermediate/mild patients survive into childhood or adulthood ([PMID: 26287655](#)).

Prognostic biomarker. Serum VLCFA, particularly **C26:0**, correlates with severity: "The best predictive value for estimating the projected disease severity and survival time is a concentration of C26:0" ([PMID: 32946460](#)).

Morbidity / function. Severe global disability — cortical blindness, deafness, no developmental milestones, seizures, hepatic and renal failure. Complications: coagulopathy, adrenal crisis, fractures, feeding failure, and vulnerability to **overwhelming neonatal sepsis** ([PMID: 33213396](#)).

Recovery potential. None for the severe form; care is palliative/supportive. Milder spectrum patients may stabilize for years ([PMID: 26287655](#)).

12. Treatment

Pharmacotherapy. Oral cholic acid (**Cholbam®**) — the **only FDA-approved therapy** (March 2015), an adjunctive treatment for ZSDs and single-enzyme bile-acid-synthesis disorders. "Cholbam® (cholic acid), approved by the U.S. Food and Drug Administration in March 2015, is currently the only therapy approved as adjunctive treatment for patients with ZSDs and single

enzyme bile acid synthesis disorders" (PMID: 34521419). It suppresses endogenous bile-acid synthesis, lowering toxic C27 intermediates and improving liver chemistries. In a Phase-3 continuation study (53 patients, 12 with ZSD), "statistically significant improvements in urinary bile acids ($P = 0.003$), height ($P < 0.001$), and body weight ($P < 0.001$) were observed" (PMID: 31899729); extension studies confirm sustained suppression (PMID: 30793331, PMID: 30519152). NCIT: C29076 (cholic acid).

Adjunct / experimental medical therapies. DHA, **Lorenzo's oil**, **batyl alcohol**, and fat-soluble-vitamin supplementation have partial/anecdotal support: "There is some support for the pharmacologic therapies of Lorenzo's oil, docosohexanoic acid, and batyl alcohol in altering symptoms; however, systematic long-term studies are lacking" (PMID: 34625341).

Advanced therapeutics (preclinical). **In vivo adenine base editing** (AAV9-ABE8e) corrected up to 60% of the pathogenic allele in liver of a Pex1-G844D ZSD mouse; "base editing eliminated bulk accumulation of very long-chain and branched-chain fatty acids, and toxic C27-bile acid intermediates," and normalized liver histology/transcriptomes (PMID: 41981313). **AAV8 gene therapy** is in development for the retinal phenotype (PMID: 42182139). **Hepatocyte transplantation** has been explored in models (PMID: 33396635).

Supportive / rehabilitative. Anti-epileptic drugs; nutritional support/gastrostomy for feeding failure; hearing amplification (improves outcomes, PMID: 34534157); low-vision support; adrenal replacement; management of coagulopathy and liver disease; physical/occupational/speech therapy. Care is multidisciplinary and largely palliative in the severe form.

13. Prevention

No primary prevention exists for this congenital, autosomal-recessive genetic disorder with no environmental/infectious cause. "As it is fatal in early life, genetic counseling and prenatal diagnosis are thus crucial" (PMID: 23327810). Preventive mainstays:

- **Genetic counseling** for at-risk couples (25% recurrence risk per pregnancy).
- **Carrier testing / cascade screening** in families and founder populations.
- **Prenatal diagnosis** on CVS/amniocytes (VLCFA / DHAPAT enzyme assay or molecular testing) and **preimplantation genetic diagnosis (PGD)**.
- **Secondary prevention:** newborn C26:0-LPC screening enables early identification and supportive intervention. Note the ethical caveat — "The Dutch Health Council recommended to screen only male newborns for ALD without identifying untreatable

conditions associated with elevated C26:0-LPC, like Zellweger spectrum disorders" (PMID: 36256460).

- **Tertiary prevention:** cholic acid, nutritional support, sensory-aid provision to limit complications.

No immunization or public-health/environmental interventions are applicable.

14. Other Species / Natural Disease

Model species with orthologs. PEX10 orthologs are conserved across eukaryotes — mouse (*Mus musculus*, NCBI Taxon 10090), zebrafish (*Danio rerio*, 7955), *Arabidopsis thaliana* (3702), and yeasts (*Hansenula/Pichia*, *Saccharomyces*). In *Arabidopsis*, PEX10 dysfunction is **embryo-lethal**: "dysfunction of a homologous gene in *Arabidopsis* leads to lethality at the heart stage of embryogenesis, impairing the biogenesis of peroxisomes, lipid bodies, and protein bodies" (PMID: 12883010), underscoring deep evolutionary conservation of the peroxisome-import machinery.

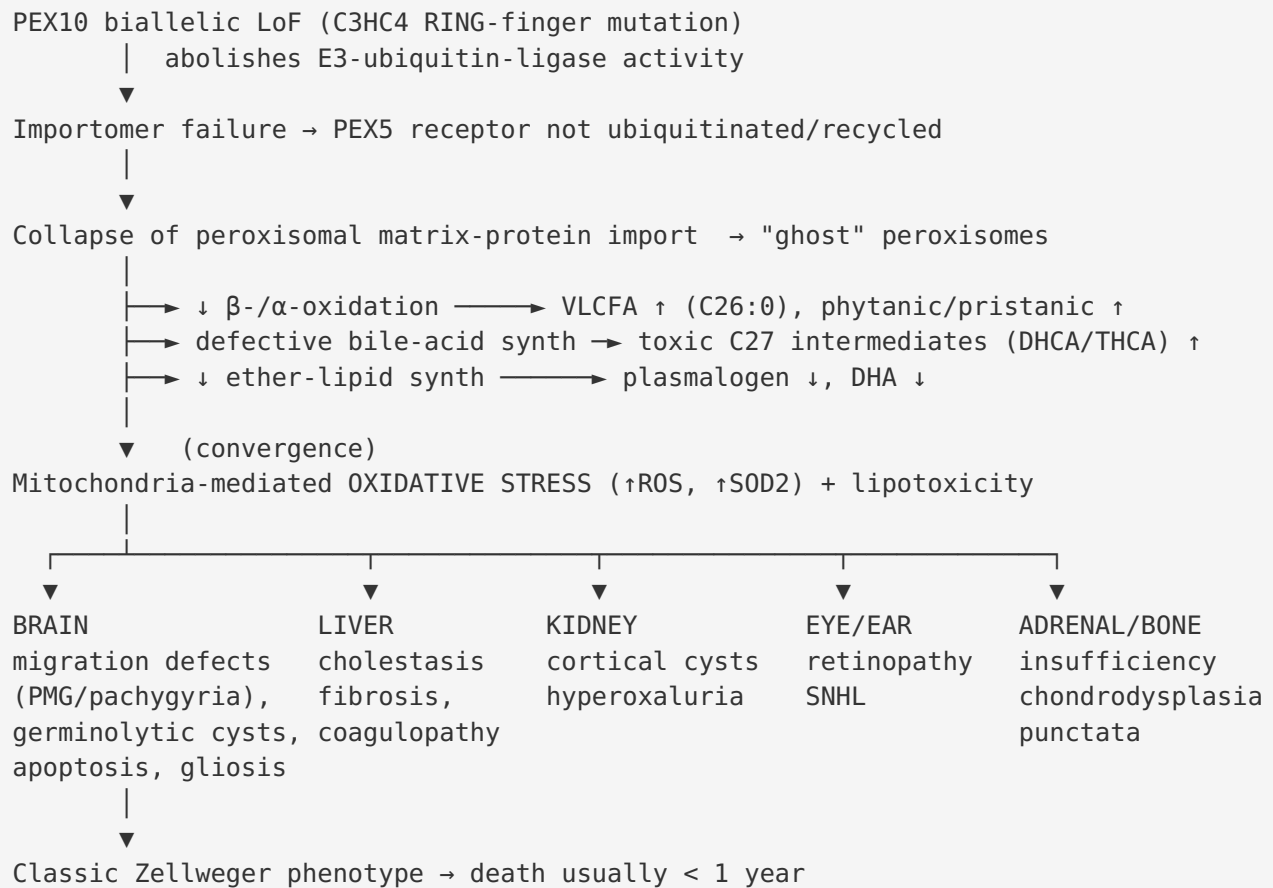
Natural disease in companion/wildlife species. No well-characterized naturally occurring PEX10-Zellweger equivalent is documented in the reviewed literature (OMIA searches were not resolved in this investigation). **Not zoonotic** — this is a Mendelian metabolic disorder, not transmissible.

15. Model Organisms

Model	Type	Key features / recapitulation	Reference
PEX1-p.Gly844Asp (G844D) mouse	Mammalian knock-in	Models the common human PEX1-p.Gly843Asp allele; reproduces retinal & RPE degeneration with subretinal inflammation, liver pathology, metabolic dysfunction	PMID: 40058592 , PMID: 41981313
Brain-restricted PEX13-deficient mouse	Mammalian conditional KO	Reduced plasmalogens, impaired cerebellar development, defective granule-cell migration, astro-/microgliosis, ↑ ROS/MnSOD, neuronal apoptosis — mechanistic Zellweger brain model	PMID: 20959636
Pex11α-KO mouse	Mammalian KO	Serum/liver/heart lipidomic, metabolomic, proteomic dysregulation	PMID: 35083512
Zebrafish pex mutants (e.g., PEX13)	Vertebrate	Established peroxisome/pexophagy models	PMID: 36541703
<i>Arabidopsis</i> AthPEX10 T-DNA mutant	Plant	Embryo-lethal; absent peroxisomes/oil bodies — proves conserved essentiality	PMID: 12883010
Yeast (<i>Pichia/Hansenula</i>)	Fungal	Defined importomer/Pex20 ubiquitination biochemistry	PMID: 23344950

Applications & limitations. Models faithfully reproduce the **biochemical lesion** (VLCFA/plasmalogen/bile-acid abnormalities) and organ pathology (retina, liver, cerebellum), and have enabled proof-of-concept **base-editing and AAV gene therapy** ([PMID: 41981313](#), [PMID: 42182139](#)). Limitations: most established models use **PEX1**, not PEX10; complete null models are often perinatally lethal, limiting study of later neurodegeneration; and no single model captures the full human multisystem severity spectrum.

Mechanistic Model / Interpretation



The unifying interpretation is that **a single upstream molecular lesion (RING-E3 failure) produces a broad metabolic derangement** because peroxisomes host many non-redundant pathways. Severity tracks with **residual PEX10 function**: null/severe alleles → Zellweger; hypomorphic alleles → NALD/ataxic form with survival into adulthood. **C26:0 is both the diagnostic and prognostic readout** of this pathway, and correcting the pathway (base editing) reverses the biochemistry in models — validating the causal chain.

Evidence Base

PMID	Contribution	Role
32069232	PEX10 among 13 PEX genes; genotype–phenotype correlation	Supports etiology (F001)
28320181	PEX10 import function; severity spectrum; RING p.C307Y	Supports gene function/variants (F001)
20679226	PEX10/PEX2/PEX12 zinc-RING E3 ligases	Supports mechanism (F012)
23344950	RING peroxins ubiquitinate PTS receptors	Supports mechanism
34625341	VLCFA/bile-acid metabolic failure; adjunct therapies	Supports mechanism/treatment (F002, F003)
28784167	Classic Zellweger neonatal phenotype	Supports phenotype (F002)
35741019	Severity-graded clinical findings (meta-analysis)	Supports phenotypes (F004)
37541626	Near-universal retinal degeneration	Supports phenotype (F004)
34534157	Sensorineural hearing-loss characterization	Supports phenotype (F004)
20959636	Oxidative-stress neurodegeneration (PEX13 brain model)	Supports mechanism (F007)
28677031	C26:0-lysoPC DBS marker (89.2% sensitivity)	Supports diagnostics (F008)
36256460	ZSD detected as secondary finding of C26:0-LPC NBS	Supports diagnostics (F008)
32946460	C26:0 best predictor of severity/survival	Supports prognosis (F009)
18415699	MRI: polymicrogyria/pachygyria, germinolytic cysts	Supports diagnostics (F009)
12069541	Historical CHRS synonym; early-death course	Supports identity/prognosis (F009, F010)
23327810	Fatal AR disorder; counseling/prenatal prevention	Supports identity/prevention (F010)
30846882	Incidence ~1/90,000 (NBS)	Supports epidemiology (F006)
22894767	PEX6 founder effect Quebec (1/12,191; carrier 1/55)	Supports epidemiology (F006)

PMID	Contribution	Role
12794690	PEX10 founder 2-bp deletion, Japan	Supports epidemiology (F006)
34521419	Cholic acid sole FDA-approved therapy	Supports treatment (F003)
31899729	Phase-3 cholic-acid efficacy (urinary BA P=0.003)	Supports treatment (F003)
41981313	In vivo base editing rescues ZSD mouse	Supports treatment/models (F005)
40058592	PEX1-G844D mouse retinal/RPE phenotype	Supports models (F005)
32904102	DBP deficiency = Zellweger-like DDx	Supports diagnostics (F012)
26387595	Heimler = mild PBD (PEX1/PEX6)	Supports DDx/spectrum
33213396	Peroxisomes in immunity; neonatal sepsis	Supports anatomy/immune (F011)
37144748	Craniofacial/neonatal presentation (PEX6 severe)	Supports anatomy (F011)
12883010	Arabidopsis PEX10 embryo-lethal; conservation	Supports models (F012)
39359950	Lists related PBD phenotypes (differential spectrum)	Supports DDx

Limitations and Knowledge Gaps

- 1. PEX10-specific data are sparse.** Much mechanistic and therapeutic evidence derives from **PEX1** (most common ZSD gene) and **PEX13** models; direct PEX10 mouse models with full multisystem recapitulation are lacking. Genotype–phenotype claims for PEX10 rest on relatively few case reports.
- 2. No PEX10-specific epidemiology.** Incidence figures (1/50,000–90,000) are pan-ZSD; the PEX10 fraction and its precise carrier frequency (outside the Japanese founder allele) are not well quantified.
- 3. Prognostic quantification is limited.** C26:0 predicts severity, but no validated multivariable survival model exists for PBD6A specifically.
- 4. Therapeutics are palliative.** Cholic acid addresses the bile-acid arm only; it does not correct VLCFA/plasmalogen deficits or CNS disease. Base editing/gene therapy remain **preclinical**, tested largely in PEX1 models and in accessible organs (liver, retina), not brain.

5. **Epigenetics, standardized QoL instruments, and veterinary/natural-disease data** were not resolved in this investigation and represent genuine gaps.
6. **Newborn-screening ethics.** C26:0-LPC screening detects untreatable ZSD as a secondary finding — a policy/counseling challenge rather than a clinical benefit.

Proposed Follow-up Experiments / Actions

1. **Generate a PEX10 RING-domain knock-in mouse** (e.g., p.C307Y) to obtain a PEX10-specific severity model and compare with PEX1-G844D.
2. **Establish PEX10 genotype–phenotype curation** across ClinVar/case literature, correlating residual E3-ligase activity (in vitro PEX5-ubiquitination assays) with clinical severity and C26:0.
3. **Extend base-editing/AAV gene therapy to PEX10 and to CNS delivery** (BBB-crossing AAV capsids; intrathecal routes), testing whether early correction prevents neuronal-migration defects (requires prenatal/perinatal timing).
4. **Prospective natural-history registry** capturing standardized QoL (infant-adapted PROMIS), survival, and biomarker trajectories to build a PBD6A-specific prognostic model.
5. **Biomarker refinement:** validate C26:0-lysoPC cutoffs and combine with bile-acid intermediates for higher specificity in newborn screening, and evaluate plasmalogen/DHA as treatment-response markers.
6. **Screen approved drugs / chaperones** that stabilize hypomorphic PEX10 or upregulate residual peroxisome import, potentially converting severe to milder phenotypes.
7. **Search OMIA / veterinary databases** to determine whether naturally occurring peroxisome-biogenesis disease exists in companion animals for comparative study.

Report compiled from a 5-iteration autonomous investigation: 12 confirmed findings, 55 papers reviewed. Evidence sources span human clinical cohorts, model-organism (mouse/zebrafish/plant/yeast) studies, in vitro biochemistry, and computational/population-genetic estimates.